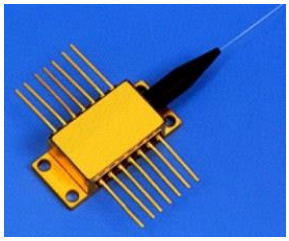
LASER DIODE FLPD-1028-50-DFB-BTF



The FLPD-1028-50-DFB-BTF laser diode is a cost effective, highly coherent laser source. The DFB laser diode chip is packaged in an industry standard hermetically sealed 14 pin butterfly package with TEC and PD.

Features

- Narrow Linewidth < 2MHz
- Excellent Wavelength Control and Stability
- Optical Isolator
- Mode-Hop Free Tuning
- Customer Specific Wavelengths Available

Applications

- Tunable Diode Laser Absorption Spectroscopy
- Seed Laser
- Fiber Laser

Absolute Maximum Ratings

Case Temperature	-5°C - +70°C
Operating Temperature	+10°C - +40°C
Forward Current	250mA
Forward Voltage	1.8V
Reverse Voltage (LD)	2.0V
Reverse Voltage (PD)	20V

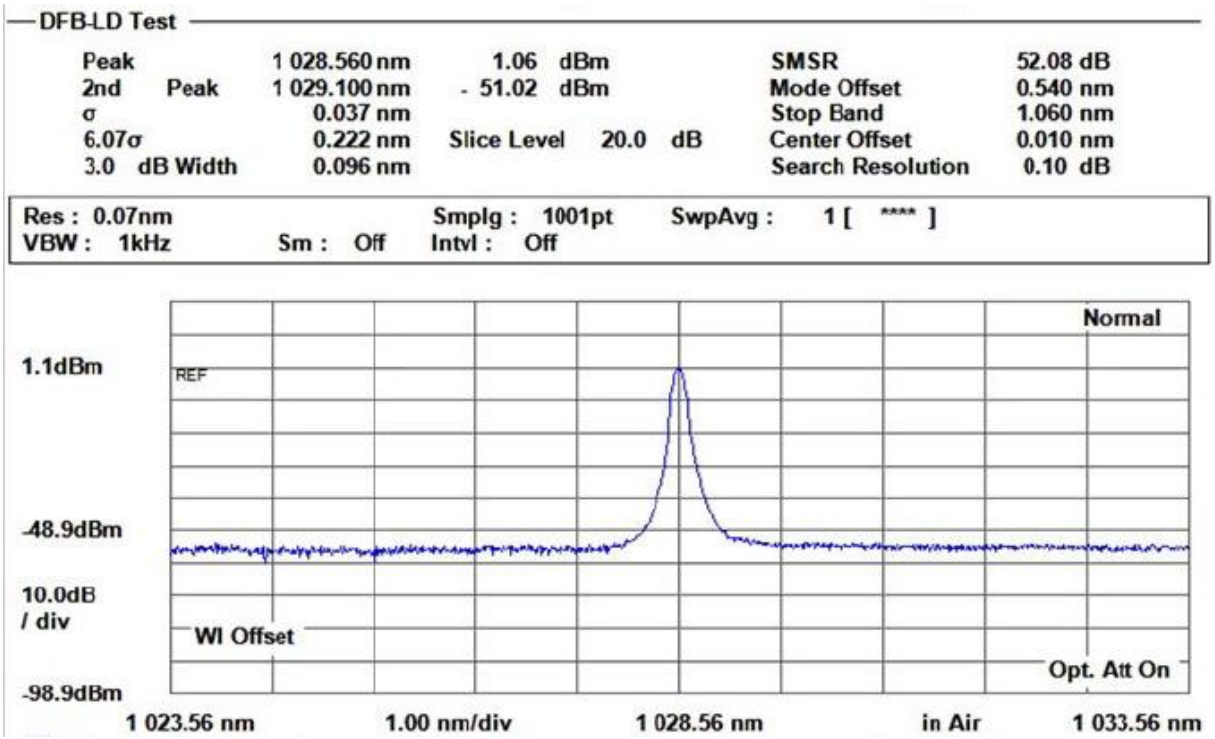
FRANKFURT LASER COMPANY
An den 30 Morgen 13, 61381 Friedrichsdorf, Germany
Tel.: +49(0)6172.27978-0 Fax: +49(0)6172.27978-10
E-mail: sales@frlaserco.com; Internet: **WWW.FRLASERCO.COM**

Electrical/Optical Characteristics

Parameter	Symbol	Units	Min	Typ	Max	Test Conditions
Emission Wavelength	λ_R	nm	1027.5	1028.0	1028.5	T=25°C
Fiber Output Power	P_{OP}	mW		50	55	T=25°C
Side Mode Suppression Ratio	SMSR	dB	30	40		T=25°C
Optical Isolation	OI	dB	30			
Threshold Current	I_{TH}	mA		47		
Laser Current	I_{OP}	mA		230	250	
Laser Voltage	U_{OP}	V		1.6	1.8	
Quantum Efficiency	η			0.18		
Current Tuning Coefficient	$\Delta\lambda/\Delta I$	nm/mA		0.015		
Temperature Tuning Coefficient	$\Delta\lambda/\Delta T$	nm/K		0.12		
TEC Current		A			1.2	Appropriate heat sink required
NTC Thermistor Resistance		KΩ	9.5	10	10.5	T=25°C
NTC Temperature Coefficient		%/°C	-4.4			
Fiber + Connector			HI1060 / PM980 Fiber* + FC/APC			

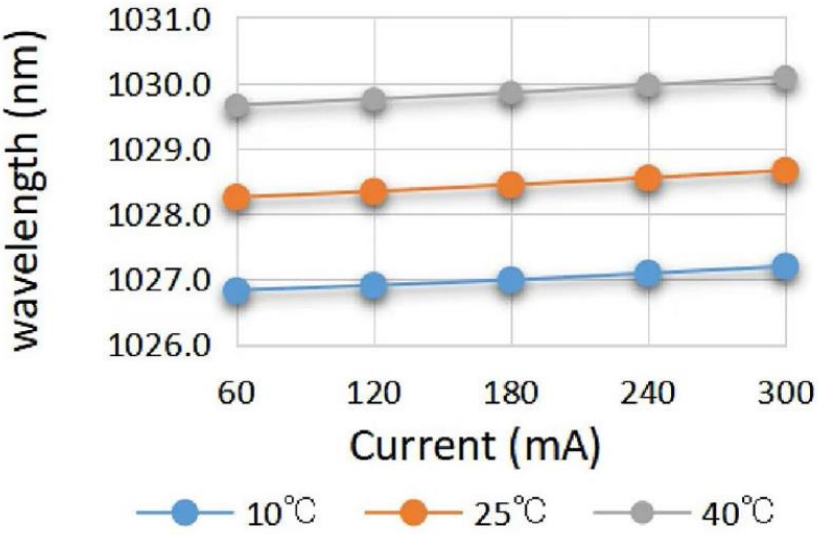
*Available with SM and PM fiber. Please specify.

Spectrum

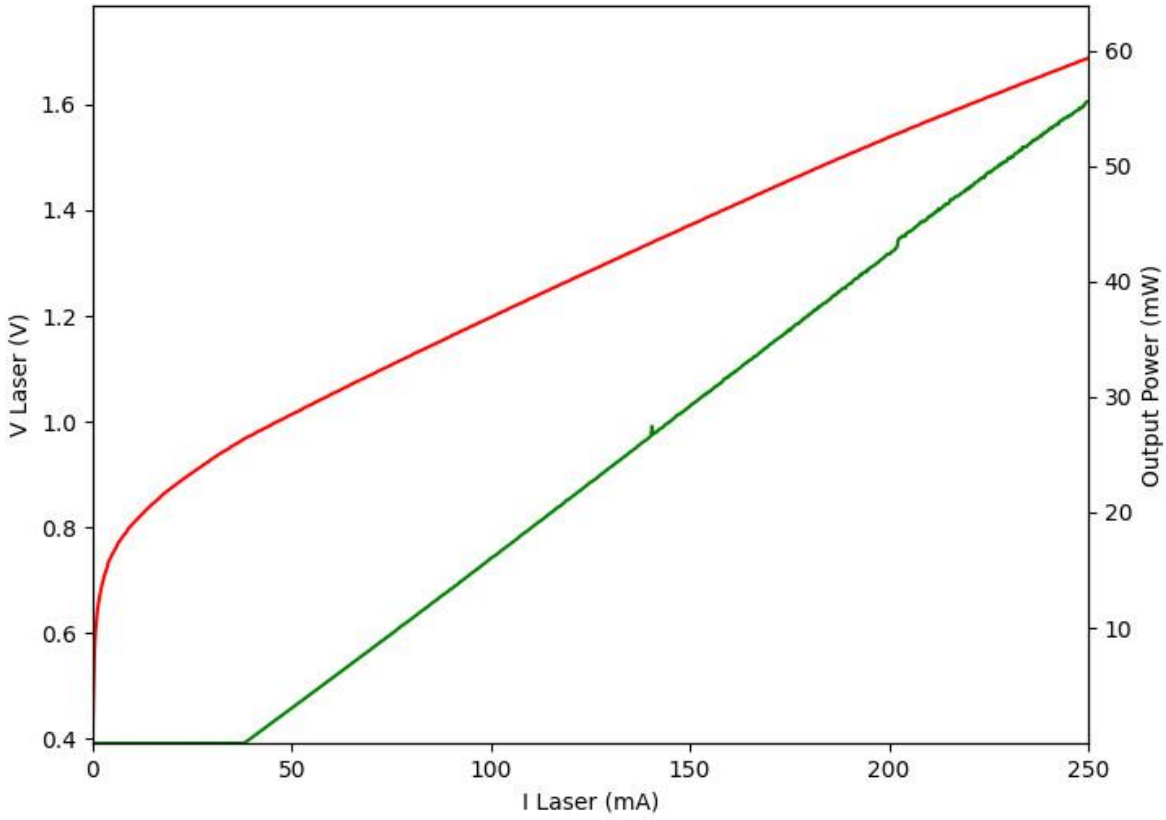


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Tuning Characteristics

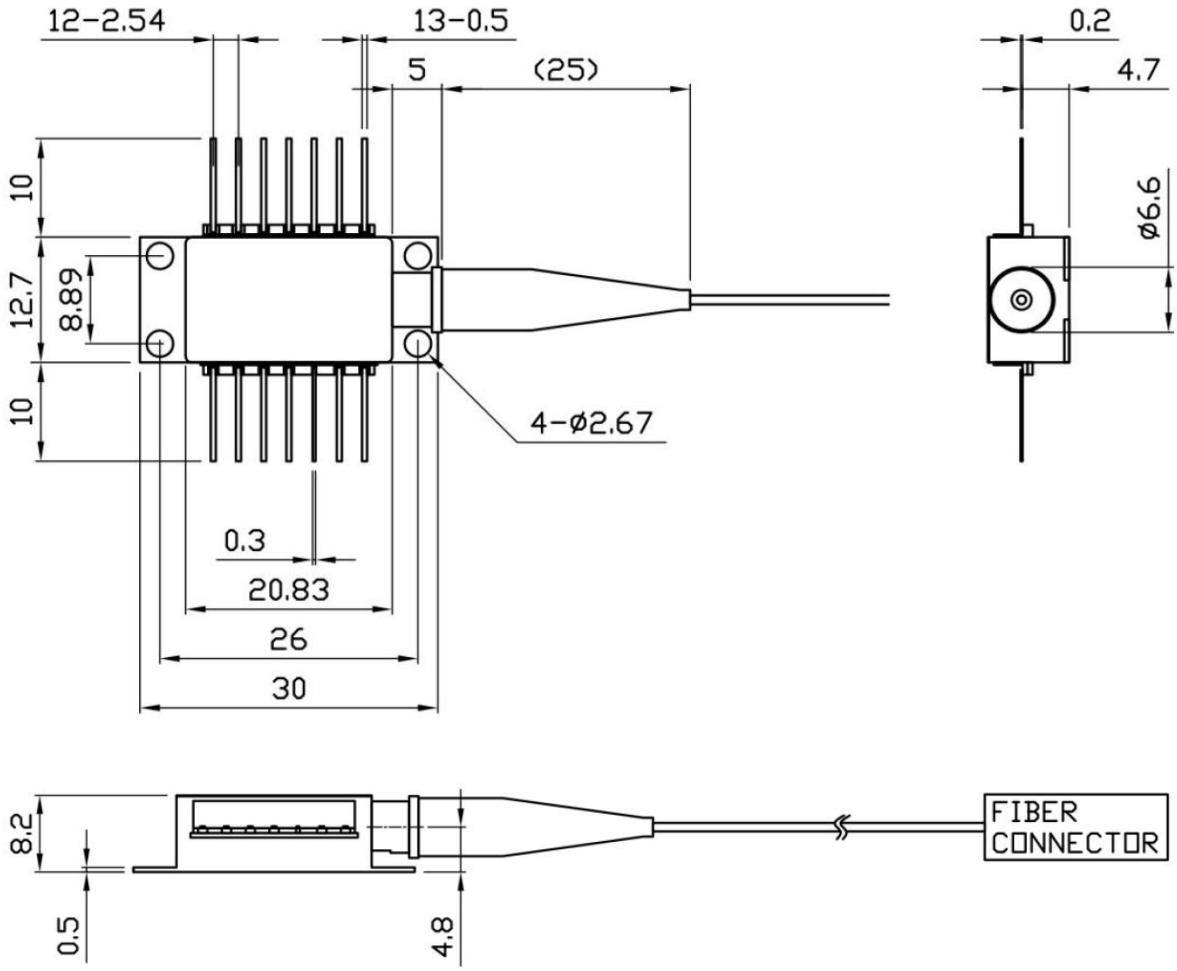


P-I & P-V Curve



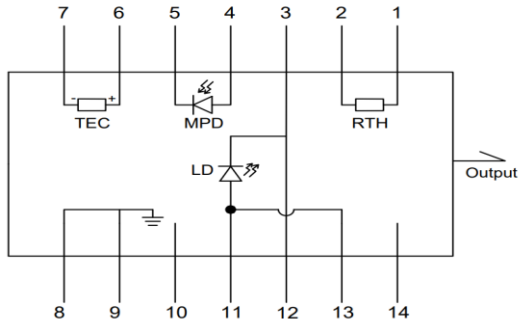
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Package Size and Pin Definition



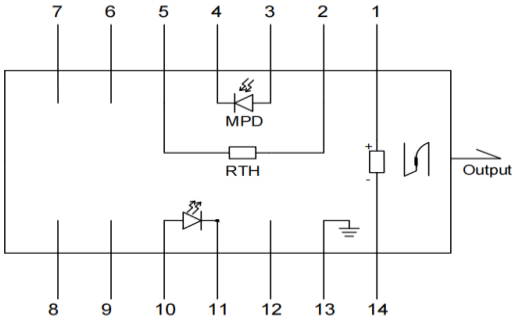
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Pin-Out (Type A)



PIN#	Function	PIN#	Function
1	Thermistor	8	Case Ground
2	Thermistor	9	Case Ground
3	LD(-)	10	NC
4	Detector (+)	11	LD(+)
5	Detector (-)	12	LD (-), RF
6	TEC (+)	13	LD(+)
7	TEC (-)	14	NC

Pin-Out (Type B)



PIN#	Function	PIN#	Function
1	TEC(+)	8	Case Ground
2	Thermistor	9	NC
3	Detector(+)	10	Laser Anode (+)
4	Detector(-)	11	Laser Anode (-)
5	Thermistor	12	NC
6	NC	13	NC
7	NC	14	TEC(-)

* Other pin-out on request.

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